

The Binary Norm Property for Cayley–Dickson Associators

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Abstract. We prove that for any Cayley–Dickson algebra of dimension 2^k ($k \geq 3$), the associator of three imaginary basis elements has norm either 0 or 2, with no intermediate values. The proof is elementary: it follows from the $\mathbb{Z}_2^k$ grading of the Cayley–Dickson sign function and the associativity of bitwise exclusive-or. As a corollary, basis non-associativity in the entire Cayley–Dickson tower is a $\mathbb{Z}_2$ parity phenomenon — a sign flip within a single basis direction. We verify the result computationally through dimension 128 ($k = 7$, classifying $127^3 = 2\,048\,383$ triples) and confirm the conjectured nonzero count $T_7 = 1\,046\,808 = 6231 \times 168$ [1].

Keywords: Cayley–Dickson algebras, associator, norm dichotomy, sign function

MSC 2020: 17A75, 17D05

1. Introduction

In a companion paper [1], we conjectured that the number of ordered triples of imaginary basis elements with nonzero associator in the 2^k -dimensional Cayley–Dickson algebra is $T_k = 168(P_k - 4P_{k-1})$, and observed computationally that all nonzero associator norms equal exactly 2 through dimension 64 ($k = 6$). The binary norm property ($\| [e_i, e_j, e_k] \| \in \{0, 2\}$) was listed as an open question.

In this note we prove the binary norm property for *all* Cayley–Dickson algebras, using an elementary argument based on the $\mathbb{Z}_2^k$ grading of the construction. The proof is dimension-independent and requires no case analysis.

2. The Cayley–Dickson sign function

Let A_k denote the Cayley–Dickson algebra of dimension 2^k , with imaginary basis elements $e_1, \dots, e_{2^{k-1}}$ indexed by nonzero vectors of $\mathbb{Z}_2^k$. The Cayley–Dickson construction determines a *sign function* $\sigma : (\mathbb{Z}_2^k \setminus \{0\})^2 \rightarrow \{+1, -1\}$ such that:

$$e_i \cdot e_j = \sigma(i, j) \cdot e_{i \oplus j} \tag{1}$$

for all nonzero $i, j \in \mathbb{Z}_2^k$, where $\oplus$ denotes bitwise exclusive-or [2]. The sign function σ is defined recursively by the Cayley–Dickson doubling; its explicit form is not needed for the proof.

The key property is that $\sigma(i, j) \in \{+1, -1\}$ for all inputs — it is a *bivalent* function. This follows from the fact that each level of the Cayley–Dickson doubling combines products with coefficients ± 1 only.

3. The theorem

Theorem. For any $k \geq 1$ and any nonzero $i, j, m \in \mathbb{Z}_2^k$:

$$\| [e_i, e_j, e_m] \| \in \{0, 2\} \quad (2)$$

Proof. Expanding the associator using Equation 1:

$$(e_i e_j) e_m = \sigma(i, j) \cdot e_{i \oplus j} \cdot e_m = \sigma(i, j) \cdot \sigma(i \oplus j, m) \cdot e_{(i \oplus j) \oplus m} \quad (3)$$

$$e_i (e_j e_m) = \sigma(j, m) \cdot e_i \cdot e_{j \oplus m} = \sigma(j, m) \cdot \sigma(i, j \oplus m) \cdot e_{i \oplus (j \oplus m)} \quad (4)$$

Since $\oplus$ is associative on $\mathbb{Z}_2^k$: $(i \oplus j) \oplus m = i \oplus (j \oplus m)$. Both parenthesisations therefore yield a scalar multiple of the *same* basis element $e_{i \oplus j \oplus m}$. Define:

$$\begin{aligned} \alpha &= \sigma(i, j) \cdot \sigma(i \oplus j, m) && \text{(left parenthesisatation sign)} \\ \beta &= \sigma(j, m) \cdot \sigma(i, j \oplus m) && \text{(right parenthesisatation sign)} \end{aligned} \quad (5)$$

Since σ is bivalent, α and β are each products of two elements of $\{+1, -1\}$, hence $\alpha, \beta \in \{+1, -1\}$.

The associator is:

$$[e_i, e_j, e_m] = (\alpha - \beta) \cdot e_{i \oplus j \oplus m} \quad (6)$$

Since $\alpha, \beta \in \{+1, -1\}$:

$$\alpha - \beta \in \{-2, 0, +2\} \quad (7)$$

Therefore $\| [e_i, e_j, e_m] \| = |\alpha - \beta| \cdot \| e_{i \oplus j \oplus m} \| = |\alpha - \beta| \cdot 1 \in \{0, 2\}$. $\square$

Remark 1. The associator is nonzero if and only if $\alpha \neq \beta$, i.e., the two parenthesisations assign *opposite signs* to the same basis element. Non-associativity in the Cayley–Dickson tower is therefore a pure $\mathbb{Z}_2$ parity phenomenon: a sign disagreement, not a directional deviation.

Remark 2. The proof uses only two properties: (i) the sign function σ is bivalent, and (ii) the index function is $\oplus$ (associative). Both hold by construction at every level of the Cayley–Dickson tower. No special properties of the octonions (alternativity, norm multiplicativity, division) are required; the result applies equally to sedenions, trigintaduonions, and all higher algebras.

4. Computational verification and T_7

As a cross-check, we implemented the recursive Cayley–Dickson sign function in the Sounio programming language [3] and verified the binary norm property exhaustively at dimensions 8, 16, 32, and 128. The sign function approach avoids constructing full Cayley–Dickson multiplication tables: each associator classification requires only four evaluations of σ (two for α , two for β), each computed by integer recursion through k levels. No floating-point arithmetic or high-dimensional arrays are needed.

At dimension 128 ($k = 7$), all $127^3 = 2\,048\,383$ ordered basis triples were classified. The nonzero count is:

$$T_7 = 1\,046\,808 = 6231 \times 168 \quad (8)$$

This is the fifth verification point for the conjectured formula $T_k = 168(P_k - 4P_{k-1})$ [1]:

k	Dim	T_k	$T_k/168$	Method
3	8	168	1	Theorem [4]
4	16	1 848	11	Exhaustive
5	32	15 960	95	Exhaustive
6	64	130 200	775	Exhaustive
7	128	1 046 808	6 231	Sign function (this work)

5. Acknowledgments

The sign function approach was implemented in Sounio, a systems programming language for epistemic computing [3]. The $T_3 = 168$ base case was established in joint work with M. Gerenutti [4].

Bibliography

1. Agourakis, D. C.: The 168 Quantum of Non-Associativity in the Cayley–Dickson Tower, (2026)
2. Baez, J. C.: The Octonions. Bulletin of the American Mathematical Society. 39, 145–205 (2002)
3. Agourakis, D. C.: Sounio: A systems programming language for epistemic computing, <https://github.com/agourakis82/sounio>
4. Agourakis, D. C., Gerenutti, M.: The 168 Count for Octonion Basis Associators and a Computational Cayley–Dickson Tower Extension. Advances in Applied Clifford Algebras. (2026)